Rufayda Sanjida

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Education

Drexel University Anticipated Graduation: June 2027
Bachelor of Science in Computer Science, Minor in Mathematics | ***GPA: 3.2*** *Philadelphia, PA*

Experience

eMoney Advisor – a Fidelity Investments Company March 2026 – Sept. 2026
Software Engineer Co-op *Radnor, PA*

- Developed an end-to-end client-facing onboarding feature using **React**, **TypeScript**, and **RESTful APIs** to help users identify and prioritize their financial goals, enabling advisors to generate tailored financial plans faster
- Refactored a **multi-tenant C#/.NET** workflow service using **dependency injection**, **feature flag gating**, and partner-scoped XML configuration to enable **zero-downtime production rollout** across multiple institutional clients
- Built a reusable React Context Provider and modal with **Yup validation** and **API-driven** state, decoupling legacy architecture across **30+ files** to eliminate prop drilling and enable independent feature access

Hologic March 2025 – Sept. 2025
R&D Engineer Co-op *Newark, DE*

- Developed computer vision software for **embedded systems** in mammography machines using **Python**, **OpenCV**, and **camera SDKs** to measure breast thickness within **±5 mm**
- Improved skin-edge detection accuracy/precision by **50%** using Gaussian filtering and Canny image-processing algorithms
- Benchmarked **Time-of-Flight** depth cameras, achieving over **98%** accuracy in distance measurement tests
- Performed **latency analysis** and **hardware-software** integration to guide next-generation imaging system design
- Collaborated with four partner companies to align testing, calibration, and prototype validation

Drexel University June 2024 – Dec. 2024
Scientific Software Developer Co-op *Philadelphia, PA*

- Built a **Python Panel-based web interface** for uploading and managing **JSON-formatted** scientific datasets, implementing authentication, file navigation, and data I/O workflows and reducing upload time by 40%
- Integrated **RESTful API** endpoints for **CRUD** operations, metadata sync, and collection management
- Automated JSON metadata parsing with Python scripts and API pipelines, reducing manual work

Projects

ReadTheRoom September 2026
Winning Hackathon Project - [MLH] Best Use of Presage *Temple University*

- Built a full-stack video analytics platform using **React 19**, **TanStack Start**, **FastAPI**, and **SQLite** that detects audience facial reactions and visualizes them on an interactive timeline with **Recharts**, deployed with **Docker Compose**
- Integrated **Google MediaPipe Tasks Vision** via **WebAssembly** to track **52 facial blendshapes** at **60 FPS**
- Built an AI chat agent using the **Google Gemini API** to query viewer sentiment and reaction timing using natural language

Finance Simulator January 2026 – Present
Personal Project *Philadelphia, PA*

- Built a **React/Node.js** web app that models long-term savings growth using compounding and **Monte Carlo simulation**, with adjustable contribution rate, rate of return, and inflation to visualize probabilistic retirement outcomes with **Recharts**
- Engineered an Express backend to run and aggregate 1,000 Monte Carlo simulations for each financial scenario

DroneDash October 2025
Winning Hackathon Project - Future Mobility Track *Temple University*

- Developed a full-stack drone delivery platform using **React**, **TailwindCSS**, **FastAPI**, and **MongoDB** with real-time delivery tracking and geospatial route optimization, leading a 5 person team to win Future Mobility track at OwlHacks 2025
- Implemented **WebRTC** communication enabling laptop-to-phone streaming, reducing demo setup by **90%**
- Built an **interactive 3D map** using Three.js and MongoDB geospatial queries to render **250+ location-based data** points

Awards and Recognition

OwlHacks 2026 Hackathon Winner: ReadTheRoom – Major League Hacking (MLH) Best Use of Presage
OwlHacks 2025 Hackathon Winner: DroneDash – Future Mobility Track
Co-Inventor, Hologic Patent-Pending Breast Imaging System – improved breast positioning and imaging coverage
Drexel Liberty Full Scholarship (Full Tuition Award) – awarded for academic excellence and community leadership

Technical Skills

Languages: Python, Java, C, C#/.NET, Bash, HTML/CSS, JavaScript, TypeScript, SQL
Web Development: Flask, FastAPI, Django, React, React Native, Node.js/Express, TailwindCSS, WebSocket, Spring Boot
AI/ML: PyTorch, Pandas, TensorFlow, OpenCV, Hugging Face, NumPy, scikit-learn
Tools: Docker, Git, Linux, Amazon Web Services (AWS), Postman, Confluence, CI/CD Pipeline, JUnit, Bitbucket, Jira